

SOFTWARE REQUIREMENTS SPECIFICATION

Task Manager Application

Version 1.0 (IEEE 830-1998 Standard Format)

Team Number	Team #4 (Section C)
Project Title	Task Manager Application
Domain / Category	Productivity, Agile Project & Workflow Management
Target Platform	Responsive Web Application (React.js + Node.js / Django)
Submission Date	October 2026

Author / Member Name	USN / SRN	Assigned Functional Modules
Chetana V Iyer	PES1UG24CS130	User Auth & RBAC; Team/Workspace Collaboration
Dhadhal Rudra	PES1UG24CS146	Task Lifecycle CRUD; Subtasks & File Attachments
Gagan B Sasalatti	PES1UG24CS164	Task Assignment & Kanban Board; Search & Filtering
Harsh Pandya	PES1UG24CS182	Progress Analytics & Reporting; Notifications & Audit Log

Revision History

Name	Date	Reason for Changes	Version
Team 4 Core Authors	02/10/2026	Initial baseline specification drafted per IEEE 830 format	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete software requirements for Version 1.0 of the **Task Manager Application**. The document details the functional capabilities, external interfaces, data structures, and quality attributes required to guide developers, testers, and academic evaluators during sprint cycles.

1.2 Intended Audience

This document is intended for: (a) Course Instructors and Faculty Evaluators for academic milestone assessment; (b) The four Student Developers (Team 4) implementing frontend, backend, and database modules; (c) Quality Assurance and Peer Testers validating test cases against requirements; (d) End users seeking clear expectations of system operations and workflows.

1.3 Product Scope

The **Task Manager Application** is an agile, multi-tenant digital productivity and workspace management platform. It empowers individuals, student cohorts, and professional teams to create shared workspaces, organize complex projects into manageable tasks and subtasks, track progress dynamically through interactive Kanban boards, execute multi-parameter search/filters, and generate insightful productivity reports with automated notifications.

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- PES University Department of CSE: Software Engineering (UE24CS341A) Mini-Project Guidelines.
- React.js & Node.js Official Engineering Specifications and Documentation.

1.5 Overview

Section 2 describes general system context, user classes, operating environment, and constraints. Section 3 outlines external interfaces and analysis models. Section 4 provides detailed functional requirements for all 8 system features. Section 5 defines non-functional criteria. Appendices present the data dictionary and Requirement Traceability Matrix.

2. Overall Description

2.1 Product Perspective

The Task Manager Application is a self-contained, web-based client-server software system. It operates via a modern single-page application (SPA) architecture communicating with a stateless RESTful backend API over HTTPS. Persistent state is maintained in a normalized relational/document database with integrated file storage for task attachments.

2.2 Product Functions

The system delivers eight central functional capabilities:

- 1. **User Authentication & RBAC:** Secure registration, JWT-based login, password recovery, and role management.
- 2. **Workspace & Team Management:** Creation of isolated workspaces, member invitation, and collaborative access control.
- 3. **Task Lifecycle Management:** Full CRUD lifecycle for tasks with priorities, deadlines, and multi-state workflows.
- 4. **Subtasks & Attachment Handling:** Granular subtask checklists with progress roll-up and multi-format document uploads.
- 5. **Kanban Workflow & Assignment Engine:** Drag-and-drop board state transitions and dynamic collaborator assignment.
- 6. **Search, Filtering & Tagging Engine:** Real-time full-text search, composite tag querying, and multi-field sorting.
- 7. **Progress Analytics & Exportable Reports:** Visual sprint completion velocity, workload distribution charts, and PDF/CSV export.
- 8. **Notification & Audit Feed:** Real-time in-app deadline alerts and immutable activity logs recording workspace mutations.

2.3 User Classes and Characteristics

User Class	Privilege Level	Key Responsibilities & Characteristics
Workspace Admin	High (Full Access)	Creates workspaces, manages team billing/settings, invites/removes members, assigns roles, and views audit logs.
Project Contributor	Medium (Read/Write)	Creates and updates assigned tasks, marks subtasks complete, uploads attachments, moves Kanban cards, and views analytics.
Guest / Viewer	Low (Read-Only)	Inspects task progress, boards, and analytics without modification rights. Suitable for clients and stakeholders.

2.4 Operating Environment

- **Client-Side:** Standard web browsers (Google Chrome 110+, Mozilla Firefox 115+, Microsoft Edge 110+, Safari 16+) with JavaScript enabled.

- **Server-Side:** Node.js runtime (v18+ LTS) or Python (3.10+) operating on Linux (Ubuntu 22.04 LTS), macOS, or Windows 10/11.
- **Database Tier:** PostgreSQL 15+ / MySQL 8.0+ or MongoDB 6.0+; Redis 7.0+ for session caching and rate-limiting.

2.5 Design and Implementation Constraints

- **RESTful Architectural Conformance:** All client-server communications must follow strict REST principles with JSON data payloads.
- **Stateless Authentication:** Secure JSON Web Tokens (JWT) signed with HMAC-SHA256 with 24-hour expiration.
- **Attachment Storage Limit:** Individual file attachments are capped at 10 MB per file, restricted to standard MIME types (PDF, PNG, JPG, DOCX, ZIP).
- **Academic Deadlines:** Software Architecture Document (SAD) and Test Plan due before October 7, 2026.

2.6 Assumptions and Dependencies

- Continuous network connectivity is assumed between client browsers and the backend application server.
- External SMTP relay service (e.g., SendGrid, Nodemailer SMTP) is assumed available for password reset and notification emails.

3. External Interface Requirements & Analysis Models

3.1 User Interfaces

The web user interface follows a modern, responsive layout adhering to WCAG 2.1 AA accessibility guidelines. Core views include: (a) Authentication Portal (Login, Registration, Password Reset); (b) Workspace Dashboard (Overview of workspaces and quick actions); (c) Interactive Kanban Board (Four visual columns: To-Do, In Progress, In Review, Done with drag-and-drop handles); (d) Task Details Drawer/Modal (Rich metadata, subtask checkboxes, file upload zone, and comment stream); (e) Analytics & Reporting View (Velocity graphs, status pie charts, and export action buttons).

3.2 Software Interfaces

- **Database Management System:** Connected via ORM/ODM (Prisma / Sequelize for Node.js, or Django ORM for Python) handling pooled connections and schema migrations.
- **Email Dispatch Service:** SMTP interface over TLS port 587 using RFC 5322 compliant message bodies.

3.3 Communications Interfaces

- **Transport Protocol:** HTTPS (TLS 1.3) utilizing AES-256-GCM encryption.
- **Real-Time Event Channel:** WebSocket (ws:// / wss://) or HTTP Server-Sent Events (SSE) for instantaneous board updates across active team sessions.

3.4 Analysis Models: Use Case Specification

The primary system interactions are structured around three core actors: *Workspace Admin*, *Project Contributor*, and *Guest*.

Use Case ID	Primary Actor	Use Case Name	Core Flow Summary
UC-01	Unregistered User	Register Account	Submit credentials, receive verification email, activate profile.

UC-02	Registered User	Authenticate / Login	Submit email/password, validate hash, issue Bearer JWT.
UC-03	Workspace Admin	Manage Workspace	Create workspace, invite members by email, configure roles.
UC-04	Contributor	Create / Modify Task	Input task details, set due date, assign tags and priority.
UC-05	Contributor	Update Kanban Status	Drag task card between status columns, trigger status audit log.
UC-06	Contributor	Manage Subtasks	Add checklist items, toggle completion, recalculate progress %.
UC-07	Any Member	Search & Filter Tasks	Apply keyword search and multi-tag filters; render filtered results.
UC-08	Admin / Contributor	Generate Report	Select date range, compute team metrics, download PDF/CSV report.

4. System Features (Functional Requirements)

In accordance with course project guidelines, functional features are partitioned equally among all 4 team members, giving each member product ownership over two distinct, comprehensive modules.

4.1 Feature Set A: Authentication & Workspace Governance (Owner: Chetana V Iyer | PES1UG24CS130)

4.1.1 Feature 1 — User Authentication & Role-Based Access Control (RBAC)

- **Description & Priority:** Provides secure onboarding, authentication, profile management, and granular permission enforcement. Priority: **High**.
- **Stimulus/Response Sequence:** User supplies email and credentials $\rightarrow$ System validates input constraints, verifies bcrypt hash, and returns an encrypted JWT $\rightarrow$ Client stores token and initializes authorized session state.

Detailed Functional Requirements:

- **REQ-1.1 (Account Registration):** The system shall allow new users to register with full name, valid email, and password satisfying complexity rules (min 8 chars, 1 uppercase, 1 digit, 1 special char).
- **REQ-1.2 (Secure Authentication):** The system shall authenticate users against salted bcrypt password hashes and issue a signed JWT upon successful credential validation.
- **REQ-1.3 (Password Recovery):** The system shall provide an automated password reset flow generating a cryptographically secure, time-limited token (expiry 15 mins) transmitted via email.

4.1.2 Feature 2 — Workspace & Team Collaboration Management

- **Description & Priority:** Enables multi-tenant team segregation, role configuration (Admin, Member, Viewer), and member invitations. Priority: **High**.
- **Stimulus/Response Sequence:** Admin inputs workspace details $\rightarrow$ System creates workspace partition $\rightarrow$ Admin issues email invitation $\rightarrow$ Invitee accepts link and gains corresponding workspace permissions.

Detailed Functional Requirements:

- **REQ-2.1 (Workspace Creation):** The system shall allow authenticated users to instantiate distinct workspaces with custom titles and team descriptions.
- **REQ-2.2 (Member Invitation & Roles):** The system shall allow Workspace Admins to invite users via email and assign explicit roles (Admin, Member, Viewer).
- **REQ-2.3 (Access Enforcement):** The system shall restrict destructive workspace mutations (settings update, deletion, member revocation) strictly to users holding the Admin role.

4.2 Feature Set B: Task Lifecycle & Content Management (Owner: Dhadhal Rudra | PES1UG24CS146)

4.2.1 Feature 3 — Task Lifecycle & Core CRUD Operations

- **Description & Priority:** Core entity creation, editing, scheduling, and deletion operations for project tasks. Priority: **High**.
- **Stimulus/Response Sequence:** User clicks 'New Task', provides title, due date, priority $\rightarrow$ System validates required fields and commits task record $\rightarrow$ Task appears in workspace views.

Detailed Functional Requirements:

- **REQ-3.1 (Task Creation & Validation):** The system shall enable authorized members to create tasks specifying title, detailed description, due date, and priority level (Low, Medium, High).
- **REQ-3.2 (Task Modification):** The system shall allow task authors and assignees to edit all task parameters, dynamically recording modification timestamps.
- **REQ-3.3 (Task Archiving & Deletion):** The system shall support soft-deletion (archiving) and permanent deletion of tasks, cascading updates to child subtasks atomically.

4.2.2 Feature 4 — Subtasks, Checklists & Attachment Handling

- **Description & Priority:** Hierarchical decomposition of complex tasks into checklists and multi-format file attachment uploads. Priority: **Medium**.
- **Stimulus/Response Sequence:** User adds subtask item to task modal $\rightarrow$ System appends checklist element $\rightarrow$ Toggling checkbox triggers automatic parent task completion percentage calculation.

Detailed Functional Requirements:

- **REQ-4.1 (Nested Subtask Management):** The system shall allow users to add, reorder, check/uncheck, and delete nested subtask checklist items under any task.
- **REQ-4.2 (Progress Percentage Calculation):** The system shall automatically compute and display task completion percentage ($\text{Progress} = \frac{\text{Completed Subtasks}}{\text{Total Subtasks}} \times 100\%$).
- **REQ-4.3 (Secure File Attachments):** The system shall accept file uploads up to 10 MB per file, validating MIME types and providing secure download links.

4.3 Feature Set C: Workflow Engine & Discovery Services (Owner: Gagan B Sasalatti | PES1UG24CS164)

4.3.1 Feature 5 — Task Assignment & Kanban Workflow Engine

- **Description & Priority:** Visual task lifecycle board facilitating drag-and-drop column transitions and collaborator assignments. Priority: **High**.
- **Stimulus/Response Sequence:** Contributor drags task card from 'In Progress' to 'Done' $\rightarrow$ System updates task status in database $\rightarrow$ Real-time event broadcasts state change to other active viewers.

Detailed Functional Requirements:

- **REQ-5.1 (Collaborator Assignment):** The system shall allow tasks to be assigned to one or more workspace members with visual avatar indicators.
- **REQ-5.2 (Interactive Kanban Board):** The system shall render tasks in four primary workflow columns: *To-Do*, *In Progress*, *In Review*, and *Done*.
- **REQ-5.3 (Drag-and-Drop State Sync):** The system shall support smooth drag-and-drop card movements between columns, updating backend state with optimistic UI feedback.

4.3.2 Feature 6 — Multi-Criteria Search, Filtering & Tagging Engine

- **Description & Priority:** High-speed search across task parameters with multi-attribute filtering and custom categorical tagging. Priority: **Medium**.
- **Stimulus/Response Sequence:** User enters search keyword and selects 'High Priority' filter $\rightarrow$ System queries indexed database $\rightarrow$ Matching task cards are filtered within 100 ms.

Detailed Functional Requirements:

- **REQ-6.1 (Full-Text Search):** The system shall provide instantaneous keyword search querying task titles, descriptions, and tag labels.
- **REQ-6.2 (Multi-Attribute Filtering):** The system shall allow simultaneous multi-criteria filtering by priority, status, assignee, and due date range.

- **REQ-6.3 (Custom Tagging & Sorting):** The system shall allow users to create colored tags and sort task lists by due date (ascending/descending) or priority level.

4.4 Feature Set D: Analytics, Reporting & Audit System (Owner: Harsh Pandya | PES1UG24CS182)

4.4.1 Feature 7 — Progress Analytics, Metrics & Exportable Reports

- **Description & Priority:** Visual dashboard charts, team productivity KPIs, velocity metrics, and formal document exports. Priority: **Medium**.
- **Stimulus/Response Sequence:** Admin navigates to Analytics tab $\rightarrow$ System aggregates completion metrics over selected sprint date window $\rightarrow$ Visual charts and CSV/PDF export options render.

Detailed Functional Requirements:

- **REQ-7.1 (KPI Dashboard):** The system shall display aggregated metrics: Total Tasks, Completed Tasks, Overdue Tasks, and Team Velocity.
- **REQ-7.2 (Visual Charting):** The system shall render dynamic status distribution charts and member workload allocation bar graphs.
- **REQ-7.3 (Report Export Engine):** The system shall enable downloading structured progress summaries in CSV and PDF formats containing complete task execution metadata.

4.4.2 Feature 8 — Notification & Activity Audit Feed

- **Description & Priority:** Proactive deadline reminders, task assignment notices, and an immutable workspace activity audit log. Priority: **Medium**.
- **Stimulus/Response Sequence:** Task due date is within 24 hours $\rightarrow$ Automated cron job triggers $\rightarrow$ In-app notification badge increments and email alert is dispatched.

Detailed Functional Requirements:

- **REQ-8.1 (In-App Notification Center):** The system shall notify users immediately upon task assignment, comment mention, or status change with read/unread tracking.
- **REQ-8.2 (Automated Deadline Alerts):** The system shall run scheduled background checks to dispatch email notifications for tasks approaching deadlines within 24 hours.
- **REQ-8.3 (Immutable Activity Audit Log):** The system shall maintain an append-only timeline documenting all workspace events (actor, action, target entity, timestamp).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **REQ-NFR-1 (Response Latency):** 95% of standard API requests (task CRUD, status update, search) shall return responses within 200 ms under normal network conditions.
- **REQ-NFR-2 (Concurrency Support):** The system architecture shall support at least 50 concurrent active users per workspace without degradation in rendering performance.

5.2 Safety Requirements

- **REQ-NFR-3 (Data Loss Prevention):** The backend database shall perform automated daily snapshot backups. Critical task deletion actions shall require explicit user confirmation dialogs.

5.3 Security Requirements

- **REQ-NFR-4 (Credential Protection):** Passwords shall be hashed using bcrypt with a work factor of at least 10 before persistence.
- **REQ-NFR-5 (Data in Transit):** All network communication shall be forced over TLS 1.3 (HTTPS) with HSTS headers enabled.
- **REQ-NFR-6 (Injection & Vulnerability Mitigation):** API endpoints shall sanitize inputs against SQL/NoSQL injection and Cross-Site Scripting (XSS); JWT tokens shall be validated on every private route.

5.4 Software Quality Attributes

- **Availability:** System shall target 99.5% uptime during evaluation and sprint cycles.
- **Maintainability:** Codebase shall follow modular architecture with comprehensive unit test coverage and clear documentation.
- **Usability:** The UI shall be intuitive, requiring zero formal training for users to create and track tasks.

5.5 Business Rules & Domain Requirements

- **Rule 1:** A user cannot be assigned to a task in a workspace they are not currently a member of.
- **Rule 2:** Deletion of a workspace automatically soft-deletes all associated tasks, subtasks, and files.
- **Rule 3:** A task cannot have a due date in the past upon initial creation.

Appendix A: Glossary

Term / Acronym	Definition
SRS	Software Requirements Specification (IEEE 830-1998 standard)
JWT	JSON Web Token: A compact, URL-safe means of representing claims between two parties
RBAC	Role-Based Access Control: Restricting system access based on assigned user roles
Kanban	Agile visual workflow framework managing work items in staged columns
CRUD	Create, Read, Update, Delete: The four basic functions of persistent storage
RTM	Requirement Traceability Matrix: A table linking requirements to design, code, and test cases

Appendix B: Field Layouts & Data Dictionary

Table B.1: Tasks Entity Field Layout

Field Name	Length	Data Type	Description	Mandatory
task_id	36	UUID	Unique primary key identifier	Yes
workspace_id	36	UUID	Foreign key referencing workspace	Yes
title	150	Varchar	Summary title of the task	Yes
description	2000	Text	Full markdown description of task	No
priority	10	Enum	Low, Medium, or High	Yes
status	20	Enum	To-Do, In-Progress, Review, Done	Yes
due_date	8	Date	Target completion deadline	No
created_by	36	UUID	Foreign key of authoring user	Yes

Appendix C: Requirement Traceability Matrix (RTM)

Cross-referencing functional requirements against design components, source code files, and test cases:

Req ID	Requirement Summary	Arch Ref	Design Ref	Code File	Unit Test ID	System Test ID
REQ-1.1	User Registration	ARC-AUTH	DSN-01	authController.js	UT-AUTH-01	ST-AUTH-01
REQ-1.2	JWT Authentication	ARC-AUTH	DSN-01	authMiddleware.js	UT-AUTH-02	ST-AUTH-02
REQ-2.1	Workspace Creation	ARC-TEAM	DSN-02	workspaceController.js	UT-WS-01	ST-WS-01
REQ-2.2	Member Roles (RBAC)	ARC-TEAM	DSN-02	rbacService.js	UT-WS-02	ST-WS-02
REQ-3.1	Task CRUD Creation	ARC-TASK	DSN-03	taskController.js	UT-TASK-01	ST-TASK-01
REQ-4.1	Subtasks Checklist	ARC-TASK	DSN-04	subtaskController.js	UT-TASK-02	ST-TASK-02
REQ-4.3	File Uploads (10MB)	ARC-FILE	DSN-04	fileUploadService.js	UT-FILE-01	ST-FILE-01
REQ-5.2	Kanban Board Columns	ARC-KANBAN	DSN-05	KanbanBoard.jsx	UT-KB-01	ST-KB-01
REQ-5.3	Drag-and-Drop Sync	ARC-KANBAN	DSN-05	taskDragDrop.js	UT-KB-02	ST-KB-02
REQ-6.1	Full-Text Search	ARC-QUERY	DSN-06	searchService.js	UT-SRCH-01	ST-SRCH-01
REQ-7.1	Analytics Dashboard	ARC-RPT	DSN-07	analyticsService.js	UT-RPT-01	ST-RPT-01
REQ-7.3	CSV/PDF Report Export	ARC-RPT	DSN-07	reportExporter.js	UT-RPT-02	ST-RPT-02
REQ-8.1	In-App Notifications	ARC-NOTIF	DSN-08	notifController.js	UT-NOTIF-01	ST-NOTIF-01
REQ-8.3	Activity Audit Log	ARC-NOTIF	DSN-08	auditLogger.js	UT-NOTIF-02	ST-NOTIF-02